

Tutorial Week 02

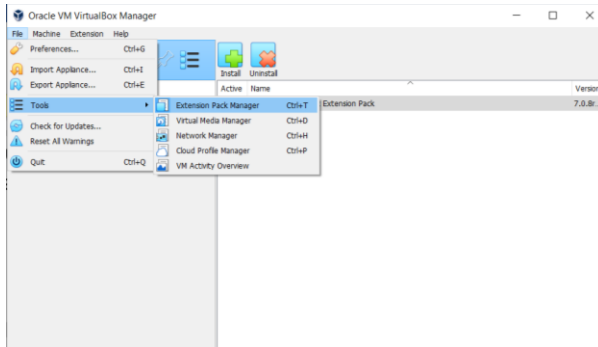
1. What are some cloud enabling technologies. Provide examples where necessary.
2. Explain connectionless packet switching, router-based interconnectivity, and transport and application layer protocols?
3. What are VDIs and how is a Guest OS different from a Host OS?
4. Explain network bandwidth and latency issues common to Internet architecture?
5. Describe data centre technology and its relevance to modern-day cloud computing?
6. Differentiate between OS and hardware-based virtualization?
7. What are KVMs and hypervisors?

Practical Lab 02**Part A**

1. Create two Virtual Machines (VMs) in Oracle Virtual Box using the Windows 10 Identical Storage Image (ISO) Operating system with the following configurations:
 - a. RAM Size: 512MB or higher
 - b. Storage: 60GB or Higher
 - c. Network Adapter: 1 x Host Only Adapter
 - d. Use ISO Image to deploy into Oracle Virtual Box
 - e. Virtual Machine Name: VM01 & VM02
2. Assign Class A static IPv4 address to VM01 & VM02. For instance, VM01 10.0.0.1 255.0.0.0 & VM02 10.0.0.2 255.0.0.0
3. Change Computer Name to SXXXXXXXVM01 & SXXXXXXXVM02. Use command prompt and verify the computer name. Screenshot the result.
4. Test connectivity from VM01 to VM02. Troubleshoot any connectivity issues. Hint: You may need to check Firewall status.
5. Join the virtual machines as a workgroup named "CLOUD".
6. Share a folder from VM01. Allow full control access to everyone. Test the shared folder from VM02.

Part B

1. Check the version of Oracle Virtual Box. For instance, version 7.0.8
2. Download the Extension Pack based on the Oracle Virtual Box version.
3. Install the Extension Pack by Selecting Tools → Extension Pack Manager and browse to the download location.



4. Once your Extension Pack matches with the Version Number, you will now be able to configure remote access to the virtual machine. Select VM01 → Settings → Display → Remote Display → Enable Server. Note down the Port Number.
5. Select Remote Desktop Connection from Physical Host and test connectivity to your virtual machine in Part 4 above. For instance, Type 127.0.0.2:Port Number. This should allow you access to remotely access your virtual machine from the Physical Host. Hint: You need the VMs running.
6. Configure remote access for VM02.

Once completed, show your results to the Facilitator.